

A 5 pillars approach to the sustainable development goals performance and reporting in Portuguese higher education institutions. Proposal for an applied framework

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Abstract

Purpose – The implementation and reporting of the sustainable development goals (SDGs) is one of the emerging challenges for higher education institutions (HEIs), but the lack of well-defined reporting structures and topics for this sector makes it difficult to map and evaluate HEI performance in relation to 2030 Agenda. This study aims to assess the SDG performance and reporting by the 13 HEIs that integrate the Times Higher Education Impact Ranking (THE_IR), from the perspective of the five pillars of the 2030 Agenda in the Portuguese context, where research on this topic is quite scarce.

Design/methodology/approach – Considering the THE_IR methodology, an SDG reporting assessment framework with 85 key topics has been developed to search for in the institutional reports of the 13 studied HEIs. For each topic, depending on the number of institutions that disclose it, a reporting index (RI) has been calculated, which was used to compare the level of reporting between the different SDGs and pillars.

Findings – Concerning HEIs' SDG performance in the THE_IR, the People pillar was the one with the most HEIs ranked, followed by the Prosperity pillar. The Planet pillar was less highlighted. SDG reporting varied widely among Portuguese HEIs. The SDG RI presented a mean value of 43.1%. The Kruskal–Wallis test revealed that the RI for the Planet pillar (RI = 23.6%) was statistically and significantly lower than those for the People and Prosperity pillars (RI = 48.6% and 53.5%, respectively). The results thus demonstrated some conformity of the SDG reporting with Portuguese HEI performance in THE_IR. Aside from Goal 5, the SDGs 4 and 3 stood out in the first four positions of the HEIs in the THE_IR. Goals 3 and 4 were also the most disclosed in institutional reports.

Originality/value – This study proposed and applied a new SDG reporting assessment framework for HEIs, contributing to better evaluation of the inclusion of important SDG-related topics in their institutional reports.

Keywords Sustainable development goals, 2030 Agenda, Pillars, Portuguese higher education institutions, Times Higher Education Impact Ranking, Performance, SDGs reporting assessment framework

Paper type Research paper



1. Introduction

In a society that is constantly changing and evolving, sustainable development has received increasing attention and concern from organizations and citizens (Blasco *et al.*, 2019). The

This work was financed by national funds through FCT – Foundation for Science and Technology, I.P., within the scope of multi-annual funding UIDB/04043/2020.

social, environmental and economic dimensions of sustainable development are universally recognized and must be synergistically contemplated in the actions of organizations (Blasco *et al.*, 2021; Pacheco *et al.*, 2020).

In 2015, the United Nations officially launched 17 sustainable development goals (SDGs), which were included in the 2030 Agenda. The three dimensions of sustainable development can be divided among five pillars (5Ps): People, Prosperity, Planet, Peace and Partnerships (United Nations, 2022).

The SDGs are designed to be universal and applied to all countries, placing education at the heart of the strategy to promote sustainable development. The contribution to the SDGs is thus only possible if there is a correct awareness, as well as the transmission of knowledge and values to companies and citizens, which is part of the core business of higher education institutions (HEIs). Among the goals of the 2030 Agenda, the goal 12.6 should be highlighted, which encourages organizations to adopt sustainable practices and integrate sustainability information into their reporting cycle (United Nations, 2022). It is thus important for HEIs to align their strategy and organizational reporting with the SDGs, informing stakeholders of their progress.

In Portugal, some initiatives and policies in the context of sustainable development have been identified in HEIs, but they are very different between institutions and have shown slow progress (Aleixo *et al.*, 2016, 2018a; Farinha *et al.*, 2018, 2019; Monteiro *et al.*, 2022b). With the launch of 2030 Agenda, the Portuguese Government (República Portuguesa, 2017) defined SDGs 4 (quality education), 5 (gender equality), 9 (industry, innovation, and infrastructure), 10 (reducing inequalities), 13 (climate action) and 14 (life under water) as strategic priorities. Given that they are key players in achieving and promoting the 2030 Agenda, it is essential to analyse HEIs' performance in this area. Some international rankings that classify HEIs' contribution to the SDGs, such as Times Higher Education Impact Ranking (THE_IR) (2022a), may represent an added value as a source of information and inspiration for the development of new structures that facilitate the evaluation of HEI performance.

This panorama creates an interesting opportunity to evaluate SDG performance and reporting in Portuguese HEIs classified in THE_IR 2022 and to answer the following research question:

RQ1. How does the performance and reporting on SDGs, and the corresponding 5Ps, vary across Portuguese HEIs?

This study therefore evaluated the SDG performance and reporting by Portuguese HEIs, from the 5P perspective. The specific objectives of the study were as follows:

- to analyse the performance of Portuguese HEIs by SDG, based on their classification in THE_IR (2022a);
- to calculate and compare the SDG reporting index (RI) and its pillars based on the information obtained from HEIs reports; and
- to verify whether the SDG RI varies according to the pillars of the 2030 Agenda.

The remainder of this paper is organized as follows: Section 2 presents the literature review; Section 3 develops the research design, including the research hypotheses, sample and research methodology; and Section 4 discusses the study's findings. The final section presents the main conclusions of the research and presents the limitations and suggestions for future research.

2. Literature review

2.1 Sustainable development goal implementation in higher education institutions

HEIs are institutions of knowledge and innovation generators with considerable influence on society. They have therefore been challenged to respond to the United Nations' call to action to incorporate the principles of the 2030 Agenda into their policies and activities (Chaleta *et al.*, 2021; De la Poza *et al.*, 2021; Leal Filho *et al.*, 2023a; Vallez *et al.*, 2022). The literature has agreed in identifying four main areas where HEIs can act in relation to the SDGs (De Iorio *et al.*, 2022; Leal Filho *et al.*, 2019; Monteiro *et al.*, 2022a; SDSN Australia/Pacific, 2017):

- (1) *Education*: the inclusion of SDGs in the curricular programmes to develop capacities and mobilize students and young people towards sustainability;
- (2) *Research*: the development of inter- and transdisciplinary research projects and works, contributing to the generation of knowledge, evidence-based solutions and innovation, which can drive the local and national implementation of the 2030 Agenda;
- (3) *Operations and governance*: alignment of the governance structure, operational policies and management of internal resources in accordance with the SDGs, including them in accountability; and
- (4) *Community engagement*: public demonstration of the commitment and mobilization of the higher education sector, facilitating intersectoral dialogue, collaboration and partnerships that strengthen actions for the 2030 Agenda.

In Portugal, studies by Aleixo *et al.* (2016, 2018a, 2018b) have confirmed the development of sustainable practices at an early stage, that is nonetheless very different among HEIs. The individual case study of the University of Évora developed by Chaleta *et al.* (2021) showed that only half of the curricular units of the university were framed with an SDG in mind. In a more generalist approach, the study by Aleixo *et al.* (2020) analysed the websites of 33 public HEIs in Portugal and revealed that only 7.7% of the courses addressed at least one SDG. In another investigation, considering the perception of 1,257 students from different Portuguese public HEIs, Aleixo *et al.* (2021) demonstrated that only half of these students knew what the SDGs are and believed that HEIs should give more training in the context of sustainable development.

The studies by Leal Filho *et al.* (2023b) with HEI members in several countries revealed that, although many organizations are already aware of the importance of sustainable development and integrate it in their institutional mission and vision, the integration of SDGs in operational processes remains limited. Thus, although there has been some progress, the existence of barriers has delayed SDG implementation in higher education; these barriers include lack of training, financial and human resources, policies and government commitment (Aleixo *et al.*, 2018b; Leal Filho *et al.*, 2019; Serafini *et al.*, 2022). Therefore, the need to accelerate efforts and public communication of performance is urgent, which makes SDG reporting a priority issue for HEIs.

2.2 Sustainable development goal reporting in higher education institutions

The growing demand for the greater accountability of and legitimacy in higher education, based on the pursuit of its “third mission” (extension activities beyond education and research), has led to the emergence of new accountability mechanisms based on the transparency of non-financial information (Brusca *et al.*, 2018; Saraite-Sariene *et al.*, 2020). It is in this context that activities within the scope of sustainable development can be framed,

as a demonstration of responsibility, commitment and credibility (Blasco *et al.*, 2019; Pacheco *et al.*, 2020). In this context, since the 1990s, several tools for evaluating and reporting the practices developed for sustainability have emerged (Sepasi *et al.*, 2018). Sustainability reports are one of the most recognized and used means for reporting such practices, including those related to the SDGs, facilitating transparency, evaluation and performance improvement, as well as stakeholder involvement (Blasco *et al.*, 2021; Caputo *et al.*, 2021; Ceulemans *et al.*, 2015).

However, despite its numerous benefits, the literature has shown that the adoption and publication of sustainability reports in HEIs are still at an early stage (Brusca *et al.*, 2018; Huber and Bassen, 2018; Sepasi *et al.*, 2018). This is not only due to the lack of institutionalization of these reports in higher education (Ceulemans *et al.*, 2015) but also due to the absence of a common, universally established and standardized reporting framework for HEIs, which hinders comparability between institutions (Caputo *et al.*, 2021; Fiorani and Di Gerio, 2022; Schoormann *et al.*, 2017). That is, in the absence of mandatory and specific guidelines, HEIs can choose to omit sustainability information, use their own tools or rely on more generalized standards.

The Global Reporting Initiative (GRI) guidelines/standards, the most widely used in the world, provide a framework and reporting indicators that allow organizations to publicly report their performance on critical sustainability issues (Di Nauta *et al.*, 2020; GRI, 2022; Huber and Bassen, 2018). However, although they can be used by HEIs, several authors have stated that the GRI guidelines are not designed and adapted to their specific needs (Fiorani and Di Gerio, 2022; Huber and Bassen, 2018; Moggi, 2019; Schoormann *et al.*, 2017). Proof of this can be found in a study by Amiano Bonatxea *et al.* (2022), who found that most European HEIs that have adopted the GRI extensively used universal standards, and only a small number of the social, economic and environmental standards were considered adequate by HEIs to disclose their activities.

According to Ribeiro *et al.* (2020), the 2030 Agenda provided new inspiration to improve the structure and harmonize sustainability reports with the specific goals stated in the agenda. In the context of higher education, Caputo *et al.* (2021) made an important contribution by conducting a content analysis of HEI sustainability reports that used the GRI standards, relating them to specific SDGs. However, even with the opportunities that the 2030 Agenda provides, the difficulties in evaluating HEI performance persist. Still little effort has been put into SDG measurement and reporting, especially in the development of common models that allow comparisons between HEIs, which has left approaches dispersed, non-systematic and fragmented (Blasco *et al.*, 2019, 2021; Caputo *et al.*, 2021; González-Torre and Suárez-Serrano, 2022).

In a study in the Spanish context, González-Torre and Suárez-Serrano (2022) developed a holistic structure, based on a disclosure index, for universities to implement in their reporting on SDGs. The results indicated different implementation and reporting methods, with different degrees of priority, adequacy, depth, reach and visibility. To address the lack of versatility of existing indicators, Griebeler *et al.* (2022) proposed a list of 61 indicators to assess the contribution of HEIs to the SDGs, while Cámara *et al.* (2021) identified a scoreboard of indicators for 12 SDGs based on a practical case from the University of the Basque Country.

However, SDG disclosure may vary according to HEI characteristics, such as the size, age and institutional diversity (De Iorio *et al.*, 2022), or the availability of financial resources and the level of internationalization (Blasco *et al.*, 2021). Regarding specific SDGs, several authors (Caputo *et al.*, 2021; Chaleta *et al.*, 2021; Leal Filho *et al.*, 2019; Monteiro *et al.*, 2022b) have pointed to the greater level of attention to SDG 4 (quality education), because it is naturally linked to the main activities of HEIs. However, the contributions of HEIs extend to all

objectives. For example, research by [Blasco et al. \(2021\)](#) on the contribution of Spanish HEIs to the SDGs has shown that health and well-being, quality education, gender equality and partnerships deserve more attention from these institutions. It is, however, more difficult for HEIs to achieve or implement active strategies related to the themes of poverty, hunger, clean water and sanitation, life under water and life on earth. With this in mind, [Daú et al. \(2023\)](#) defended the prominent position of HEIs in the objectives of quality education and reduction of inequalities.

In general, the literature has shown less involvement and/or disclosure of themes related to the environment or to social or economic issues, as demonstrated, for example, in the studies of [Aleixo et al. \(2018a\)](#) in Portuguese HEIs, [Pacheco et al. \(2020\)](#) in Ecuadorian universities and [De Iorio et al. \(2022\)](#) and [Leal Filho et al. \(2021\)](#) in HEIs from different countries and continents. Because performance assessment tools adapted to HEIs are necessary, it is important to design alternative structures, such as rankings, that can measure the contribution of HEIs to the SDGs and solve methodological problems ([Bautista-Puig et al., 2022](#)).

2.3 Times Higher Education Impact Ranking

The THE_IR assesses and rates the overall performance of HEIs in relation to the SDGs in four main areas: teaching, research, administration and outreach [[Times Higher Education Impact Ranking \(THE_IR\), 2022b](#)].

This ranking has a specific approach to the 2030 Agenda, joining others already recognized in the scope of higher education sustainability, such as the [UI Green Metric World University Ranking \(2022\)](#) or [QS World University Ranking \(2022\)](#). The importance of the THE_IR in promoting the implementation of the SDGs has been highlighted by some authors. For example, [Leal Filho et al. \(2021\)](#) pointed out that THE_IR is based on evidence of HEI success, linking practices to institutional reputation. In another perspective, [De la Poza et al. \(2021\)](#) argued that these rankings help position HEIs internationally and that the amount and variety of information that HEIs report to THE_IR demonstrate their transparency in institutional reporting. In this follow-up, the results of the THE_IR have been the target of empirical research, as in the study by [Blasco et al. \(2021\)](#), which used this ranking to measure the contribution of Spanish HEIs to the SDGs. In a more generalized approach, [De la Poza et al. \(2021\)](#) modelled and quantified the impact of the disclosure of SDG information on the score obtained by universities in the THE_IR.

THE_IR follows a methodology with several metrics and indicators for each SDG, based on research categories, continuous (measure values) and evidence (existence of policies and initiatives). HEIs can provide data for whichever SDGs they want and are subsequently classified (from 0 to 100) in the individual rankings of these SDGs. To be included in the overall ranking, only four SDGs are considered when developing the final ranking score. The submission of information related to SDG 17 (partnerships) is mandatory, and the remaining three SDGs are those with higher scores for each HEI ([Bautista-Puig et al., 2022; THE_IR, 2022b](#)). However, despite its advantages, [Bautista-Puig et al. \(2022\)](#) have pointed out some inconsistencies in the THE_IR methodology, such as the different number of indicators per SDG or the fact that the final score is based on only four SDGs, comparing and mixing HEIs with contributions to different SDGs. It is therefore necessary to develop complementary methods of performance analysis that balance and focus on all SDGs.

3. Method

3.1 Research design

The literature about SDG implementation and reporting in HEIs has shown how the lack of disclosure or differentiated reporting practices, resulting from the lack of specific guidelines

and topics for HEI, create problems in assessing and comparing HEI performance (Blasco *et al.*, 2021; Caputo *et al.*, 2021; González-Torre and Suárez-Serrano, 2022; Schoormann *et al.*, 2017). In Portugal, there has been no significant progress on this issue. Therefore, our research question is: How does the performance and reporting on SDGs, and the corresponding 5Ps, vary across Portuguese HEIs?

To our knowledge, existing studies have not specifically evaluated the performance and reporting of HEIs from the 5P perspective, which results in a new research opportunity. In this context, there is a need to develop new methodologies that demonstrate how HEIs are reporting information. The THE_IR (2022b) methodology can make an important contribution to this topic, acting as a reference framework for the development of a new reporting assessment structure. In addition, the aggregation of the SDGs around the five central axes of the 2030 Agenda, the 5Ps, allows for a more effective assessment of the current situation. Figure 1 summarizes the research design proposed for this study, which will be detailed in the following sections.

3.2 Sample

In line with other studies that focus on HEIs included in the THE_IR (Blasco *et al.*, 2021; Di Nauta *et al.*, 2020), this study used a sample of 13 Portuguese HEIs classified in the general THE_IR (2022a). According to Table 1, and considering the information of the General

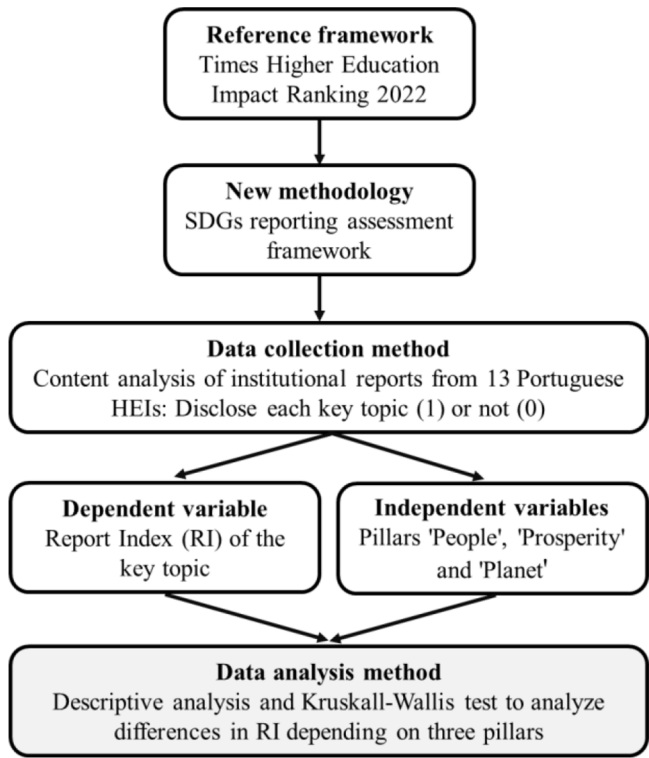


Figure 1.
Research design

Source: Author's own creation

IJSHE 25,1 110 Table 1. Sample characterization	Characteristic	N	%
	<i>Type of education</i>		
	University	10	76.9
	Polytechnic	3	23.0
	<i>Nature</i>		
	Public	12	92.3
	Private	1	7.7
	<i>Dimension</i>		
	Small ($\leq 1,000$ students)	0	0
	Medium (1,001–5,000)	2	15.4
	Large ($> 5,000$)	11	84.6
	Source: Own elaboration based on DGEEC (2022) and THE_IR (2022a)		

Directorate of Education and Science Statistics ([Direção-Geral de Estatísticas da Educação e Ciência, 2022](#)), most of these institutions are universities (76.9%), belong to the public sector (92.3%) and are large (84.5%).

In the [THE_IR \(2022b\)](#) methodology, the data used for the institution’s evaluation refer to the closest school year from January to December 2020 (except in some cases, where the data refer to 2019 due to the constraints of the pandemic). Therefore, for each of the 13 HEIs, the main institutional reports for the year 2020 were collected, depending on its public availability on the HEI website. Specifically, according to [Table 2](#), the authors looked at activity reports (6 HEIs), the activity and accounts reports (3), the management and accounts reports (2), an integrated activity and management report (1) and a quality report (1).

3.3 Sustainable development goals reporting assessment framework

A qualitative methodology was adopted through content analysis of the institutional reports. This method has been widely used in SDG reporting studies in higher education ([Aleixo et al., 2020](#); [De La Poza et al., 2021](#); [Nardo et al., 2021](#)). The content analysis technique consists of qualitative information coding related to the SDGs in predefined categories/ topics ([Krippendorff, 2013](#)). In a study applied to the Italian context, [Nardo et al. \(2021\)](#) developed a set of research keywords in HEI strategic plans, and the number of keywords varies according to different SDGs. In their critical analysis of the THE_IR methodology, [Bautista-Puig et al. \(2022\)](#) warned of a lack of homogeneity in the number of indicators per SDG, which allows the existence of metrics with different weights in the final score.

Table 2. Type of institutional reports analysed	Type	<i>n</i>	<i>n (%)</i>
	Activity reports	6	46.2
	Activity and accounts reports	3	23.0
	Management and accounts reports	2	15.4
	Integrated activity and management report	1	7.7
	Quality report	1	7.7
	Total	13	100
	Source: Institutional websites of Portuguese HEIs present in THE_IR (2022a) [1]		

Thus, to avoid biases in the level of reporting for each SDG, a reporting assessment framework has been proposed, which allows equitable consideration of all SDGs. Thus, the same number of key topics (five each) per SDG has been defined. To define these topics, the methodology of [Nardo et al. \(2021\)](#) was considered, but especially the main keywords associated with an SDG in the context of an HEI according to the [THE_IR \(2022b\)](#) methodology. As stated by [Veidemane \(2022\)](#), the creation of internationally comparable but locally significant indicators enhances the evaluation of progress and encourages HEIs to learn from each other.

In the process of analysing the reports, the keywords of a given topic were searched, not complete sentences, to cover different ways of disclosing the same information. There was some flexibility in the acceptance of expressions with equal or similar meaning in the same topic to better control the subjectivity and diversity of information. For example, the phrases “eradicate poverty” or “end poverty” (SDG 1) have the same meaning, so they will be framed in a single topic. The same is true, for example, for the use of “renewable energies” and “sustainable energies” (SDG 7), or “biodiversity conservation” and “combating the loss of biodiversity” (SDG 15). Thus, all possible terms included in the same topic were searched in each report to verify that at least one is disclosed.

In other cases, the topics may address the same subject, but do not mean the same thing, such as promoting “inclusion” and “accessible facilities for people with special needs” (SDG 10) or promoting “food security” and “purchasing from sustainable sources” (SDG 2). These terms, while related, may belong to the same SDG but not the same topic, allowing two encodings. The SDG reporting assessment framework is presented in the [Appendix](#).

In total, the 85 analysis topics in each of the 13 HEIs resulted in a total of 1,105 observations in a double check analysis method: horizontal approach, where each topic was analysed simultaneously between the 13 HEIs; and vertical approach, where all topics were analysed simultaneously in one report at a time. Data were collected between June and July 2022 and were analysed in Statistical Package for the Social Sciences software version 28.0.

3.4 Variables

The adoption of an SDG score or RI has been a data analysis method used in reporting studies ([Izzo et al., 2020](#); [Pizzi et al., 2021](#)) to quantify and frame disclosure between 0% and 100%. In this study, after classifying the absence or presence of each of the key topics in each report, a report index for each key topic (*top_RI*) was calculated, which was obtained by the number of HEIs that disclose this topic compared to the total number of HEIs (13), as shown in [Table 3](#).

Thus, each HEI can have a maximum value of 5 for each SDG (one for each key topic, and 85 in total); in turn, each key topic can have a maximum value of 13 (one for each HEI). So, following [Izzo et al. \(2020\)](#), the closer the *top_RI* is to 100%, the more institutions disclose this topic. Thus, *top_RI* is our study-dependent variable, classified as quantitative. The

$top_RI_i = \frac{\sum_{n=1}^j Topi_j}{13} \times 100$	$\frac{top_RI_i}{Top\ i}$	Report index of the key topic <i>i</i> (%) Key topic <i>i</i> under analysis, in HEI <i>j</i> . Dichotomous variable with value 1 if the HEI discloses information on this key topic and 0 otherwise
	$\frac{i}{J}$	Maximum number of key topics per HEI (85) HEIs in study (13)

Source: Author’s own creation

Table 3.
Formula of top_RI calculation

formula will subsequently be used to calculate the following: the SDG report index (SDG_RI), through the mean of the five top_RI belonging to each SDG, and the pillar report index (Pillar_RI), through the mean of the SDG_RI belonging to each pillar.

The pillars of the 2030 Agenda correspond to our independent variable as a qualitative variable. The study of Tremblay *et al.* (2020) has been followed to frame the SDGs in the five pillars of the 2030 Agenda: SDGs 1, 2, 3, 4 and 5 in “People”; SDGs 7, 8, 9, 10 and 11 in “Prosperity”; SDGs 6, 12, 13, 14 and 15 in “Planet”; SDG 16 in “Peace”; and SDG 17 in “Partnerships”.

Because the pillars Peace and Partnerships include only one SDG (16 and 17, respectively), covering only five key topics each, these two pillars were excluded from the analysis because they represented a very small number of observations. Thus, the 75 top_RI will be considered, belonging to the pillars People (Code = 1), Prosperity (code = 2) and Planet (code = 3), each concerning five SDGs and, consequently, 25 key topics.

4. Results

4.1 Portuguese higher education institution performance in Times Higher Education Impact Ranking

The first specific objective was to analyse the SDG performance of Portuguese HEIs included, in general, THE_IR 2022. The number of HEIs classified in each SDG and ranked in the top four positions with that SDG, as well as their percentage against the total number of HEIs classified in that SDG, is presented in Table 4. SDG 17 is mandatory, which is why all 13 HEIs were ranked in this goal.

Along with SDG 17, all 13 HEIs are classified for SDG 4, for which 38.5% have this goal in their top four positions. It is noted that SDGs 3 and 5, each with 12 HEIs ranked, have the highest rates of global prominence (for 41.7% and 66.7% of HEIs), which makes People the pillar with the best performance for HEIs. In turn, SDGs 1, with nine HEIs, and 2, with seven HEIs, have received less emphasis than the previous ones (11.1% and 28.6%, respectively).

Pillars	SDG	HEIs no.	No. of HEIs with the SDGs ranked in the four best positions	%
People	1	9	1	11.1
	2	7	2	28.6
	3	12	5	41.7
	4	13	5	38.5
	5	12	8	66.7
Prosperity	7	6	1	16.7
	8	9	2	22.2
	9	9	3	33.3
	10	10	1	10.0
	11	10	3	30.0
Planet	6	6	1	16.7
	12	8	1	12.5
	13	7	1	14.3
	14	6	0	0.0
	15	6	1	16.7
Peace	16	10	3	30.0
	17	13	13	100.0

Table 4.
Portuguese HEIs
performance in
THE_IR 2022

Source: Own elaboration based on THE_IR (2022a)

There seems to be a greater balance in the performance of HEIs in the Prosperity pillar. SDG 7 is the least highlighted, because it is present in the top four positions for only one (16.7%) of the corresponding six HEIs. With nine HEIs, SDGs 8 and 9 are better positioned for 22.2% and 33.3% of HEIs, respectively. Regarding SDGs 10 and 11, despite both having 10 HEIs classified, the second is more prominent (30.0%) than the first (10.0%).

The Planet pillar includes, globally, the SDGs less evident in the performance of HEIs. SDG 14 does not stand out for any of the six HEIs that correspond to it. The remaining SDGs – 6, 12, 13 and 15 – are only in the four best positions for one HEI each, making a ratio of 16.7% (of the six ranked HEI), 12.5% (eight), 14.3% (seven) and 16.7% (six), respectively.

Finally, in the Peace pillar, which concerns only SDG 16, there are 10 HEIs classified, and 30.0% of these have this SDG classified in the top four positions.

4.2 Sustainable development goal reporting and the five pillars

According to the second objective of the investigation, Figure 2 presents the SDG RI (SDG_RI) after analysing the institutional reports. It appears that SDGs 3 (SDG₃_RI = 78.5%) and 4 (SDG₄_RI = 75.4%) are the most disclosed in the reports, which both belong to the People pillar. SDG 16 (Peace pillar) had the third highest level of reporting (SDG₁₆_RI = 64.6%), while SDGs 13 (SDG₁₃_RI = 15.4%) and 14 (SDG₁₄_RI = 6.2%), both of which belong to the Planet pillar, stood out for lower levels of reporting. SDG 2 (SDG₂_RI = 15.4%) in the People pillar also stood out among the least reported. There is a greater balance of disclosure in the Prosperity pillar, with SDG_RI ranging between 43.1% (SDG 10) and 63.1% (SDG 9). In the Planet pillar, except for SDG 12 (SDG₁₂_RI = 46.2%), there is a level of disclosure below 27% for the rest. Finally, SDG 17 (SDG₁₇_RI = 38.5%) was among the least disclosed.



Figure 2.
Reporting index by
SDG

Source: Own elaboration based on United Nations (2022)

Table 5 aggregates the information regarding the pillar RI (pillar_RI), which demonstrates that Peace was the most reported (RI = 64.6%). Considering the pillars with more than one SDG, the most frequently reported was Prosperity (RI = 53.9%), but with great individual variation between HEIs (between 8.0% and 88.0%). The People pillar had a lower pillar_RI (RI = 48.6%), but with less variation between HEIs (between 24.0% and 64.0%). The Planet pillar had the lowest pillar_RI (23.6%) and with a large discrepancy between HEIs (between 0.0% and 60.0%). Finally, Partnership pillar was the second least disclosed (RI = 38.5%). Thus, considering the weight of all pillars, the 13 Portuguese HEIs had a mean RI of 43.1%, where the highest overall RI observed among HEI was 68.2% and the lowest was 14.1%.

To pursue the third objective, statistical tests were applied to verify whether the SDG RI varied according to the pillars of the 2030 Agenda. Because the Planet pillar did not have a normal distribution ($\alpha < 0.05$), a non-parametric test was performed to verify if there were statistically significant differences in RI between the People, Prosperity and Planet pillars. The Kruskal–Wallis test was used, which is the appropriate and common test in studies whose objective is to analyse differences between more than two groups (Aleixo *et al.*, 2018a, 2021).

According to Table 6, the test results ($\alpha < 0.001$) indicate that there is at least one group for which pillar_RI differs. Using group-by-group comparisons, the pillars where RI is statistically different are between People–Planet ($\alpha = 0.010 < \alpha = 0.05$) and between Prosperity–Planet ($\alpha = 0.001 < \alpha = 0.05$). The topics belonging to the pillars People (RI =

HEI*	People (N = 25) (%)	Prosperity (N = 25) (%)	Planet (N = 25) (%)	Peace (N = 5) (%)	Partnerships (N = 5) (%)	Mean** (%)
1	64.0	72.0	56.0	60.0	40.0	62.4
2	44.0	44.0	8.0	80.0	40.0	35.3
3	28.0	16.0	0.0	20.0	20.0	15.3
4	60.0	44.0	16.0	100.0	20.0	42.4
5	52.0	64.0	28.0	60.0	80.0	50.6
6	36.0	48.0	12.0	40.0	40.0	32.9
7	36.0	40.0	12.0	80.0	20.0	31.8
8	60.0	88.0	60.0	60.0	60.0	68.2
9	24.0	8.0	4.0	20.0	40.0	14.1
10	64.0	76.0	40.0	100.0	20.0	60.0
11	48.0	64.0	32.0	100.0	60.0	51.8
12	64.0	80.0	40.0	80.0	20.0	60.0
13	52.0	52.0	0.0	40.0	40.0	35.3
RI_pillar	48.6	53.5	23.7	64.6	38.5	43.1

Table 5.
Reporting index by
pillar

Notes: *Random order. **Obtained by the ratio between the number of key topics that each HEI discloses by the total number of key topics (85)
Source: Author’s own creation

Kruskall–Wallis test	p-value	Group by group	Test statistics	Adjusted p-value
15.245	$\alpha = 0.000489$	People–Prosperity	–4.640	1.000
		People–Planet	18.020	0.010
		Prosperity–Planet	22.660	0.001

Table 6.
Results of the
Kruskall–Wallis test

Source: Author’s own creation

48.6%) and Prosperity (RI = 53.6%) have a significantly higher RI than the topics for Planet (RI = 23.6%) (Table 5).

5. Discussion

This study provides a snapshot of the state of SDG implementation and reporting in Portuguese HEIs and compares it to other contexts. Firstly, the prioritization of SDGs by Portuguese HEIs in the THE_IR is similar to that of Spanish HEIs, according to [Blasco *et al.* \(2021\)](#), where the SDGs 3, 4 and 5 also stood out, and less attention was given to SDGs 6, 14 and 15.

The results of the analysis of the reports are generally compatible with the performance of Portuguese HEIs in the THE_IR. The greater attention to SDG 4 is not surprising, because it naturally represents the “first mission” of an HEI ([Daú *et al.*, 2023](#); [Monteiro *et al.*, 2022b](#)). The presence of the greatest quantity of evidence for SDG 3 may also be justified by the fact that the data corresponded to 2020, a year strongly marked by the beginning of the COVID-19 pandemic and consequent greater concern with health issues. Regarding SDG 5, its notorious prominence in the THE_IR does not seem to be so evident in the institutional reports (SDG₅_RI = 30.8%), which represents an opportunity to improve HEI disclosure on key topics of gender equality. Still, in the pillar People, it is clear that HEIs can improve the performance and dissemination of SDGs 1 and 2.

In the results, the SDGs belonging to the Prosperity pillar were those that presented the highest balance of performance in the THE_IR and disclosure in the reports. Topics related to working conditions and economic value creation (SDG 8), innovation and infrastructure (SDG 9), as well as contributions to the construction of sustainable cities and communities (SDG 11) were generally reported frequently. SDGs 7 and 10 were less evident than the others. More focus can thus be given on any of these themes, especially regarding the reduction of inequalities (SDG₁₀_RI = 43.1%), one of the priority issues for Portugal ([República Portuguesa, 2017](#)) and one of the main areas of action among HEIs ([Daú *et al.*, 2023](#)).

Both in the THE_IR and analysis of the reports, there was evidence of lower effort on the Planet pillar. This lack of attention to environmental challenges has also been evidenced in other works ([Aleixo *et al.*, 2018a](#); [De Iorio *et al.*, 2022](#); [Leal Filho *et al.*, 2021](#); [Nardo *et al.*, 2021](#); [Pacheco *et al.*, 2020](#)). This may be because the main activities of HEIs are more oriented towards education, social development, innovation and economic growth, and these themes addressed in other pillars. Although more concern is required with environmental issues, SDGs 13 and 14 were highlighted as being priorities for Portugal ([República Portuguesa, 2017](#)) and especially critical for a country with a large maritime territory. Finally, there was a positive note for SDG 16 as being moderately common among reports, unlike SDG 17, which needs to be further addressed, given the importance of partnerships evidenced in the THE_IR analysis as well.

The results of the analysis of the reports, with RI (mean) is 43.1%, demonstrate the great potential for general improvement in SDG involvement and reporting by Portuguese HEIs. The high dispersion in general RI and between pillars is a problem that illustrates the absence of SDG reporting guidelines for HEIs, as recognized by several authors ([Blasco *et al.*, 2021](#); [Fiorani and Di Gerio, 2022](#); [Griebeler *et al.*, 2022](#)). [González-Torre and Suárez-Serrano \(2022\)](#) also concluded something similar, based on quite different methods of implementation and reporting among Spanish HEIs. As [Moggi \(2019\)](#) stated, the lack of laws and reporting standards has led to the development of different documents at the level of structure, depth of disclosure and compliance.

Therefore, even if HEIs have potentially sustainable policies and practices, the lack of a system and communication strategy aligned with the themes of the 2030 Agenda hinders the way this information is disclosed and processed externally. Thus, whenever we suggest

improvements and more attention to certain themes, we refer to the way HEIs identify the key topics of disclosure, organize the information and ensure that its performance is clearly disclosed in the reports and aligned with the information sent to THE_IR.

Considering that the main barriers to the adoption of the 2030 Agenda in higher education are at the governmental, management and resource organization level (Aleixo *et al.*, 2018b; Serafini *et al.*, 2022; Leal Filho *et al.*, 2019), we believe that the implementation of a common reporting structure for the 17 SDGs and 5Ps, focusing on the four axes of action of HEIs (education, research, operations and governance and community engagement), should follow a set of steps: This begins with the establishment of governmental and institutional commitment, not only among government bodies but also among staff, teachers and researchers who contribute to the definition of priorities and strategies, according to the internal and external context in which the institution operates. The consequent planning and allocation of resources (human, financial, time and material) will serve as the basis for the implementation of initiatives and practices linked to the SDGs, whose performance should be analysed and reported to stakeholders through indicators that serve as a basis for the process of continuous improvement (Leal Filho *et al.*, 2021). The proposal of the 5P approach for SDGs implementation and reporting is illustrated in Figure 3.

6. Conclusions

The 2030 Agenda requires that HEIs are prepared to address the various challenges of sustainable development and strengthen their reporting to provide greater knowledge and external visibility of their SDG performance. Considering the expected difficulties that HEIs could face in this context because of a theoretical–practical gap with the lack of specific and mandatory reporting structures for HEIs, this study sought to answer the research question: How does the performance and reporting on SDGs, and the corresponding 5Ps, vary across Portuguese HEIs? The results confirmed the disparity in the dissemination of information, a problem evidenced not only in the asymmetries in the general reporting and 5Ps, but also in relation to the specific SDGs, to the detriment of environmental objectives. Therefore, these findings are in accordance with prior research (Aleixo *et al.*, 2018a; Blasco *et al.*, 2021; Caputo *et al.*, 2021; Leal Filho *et al.*, 2021).

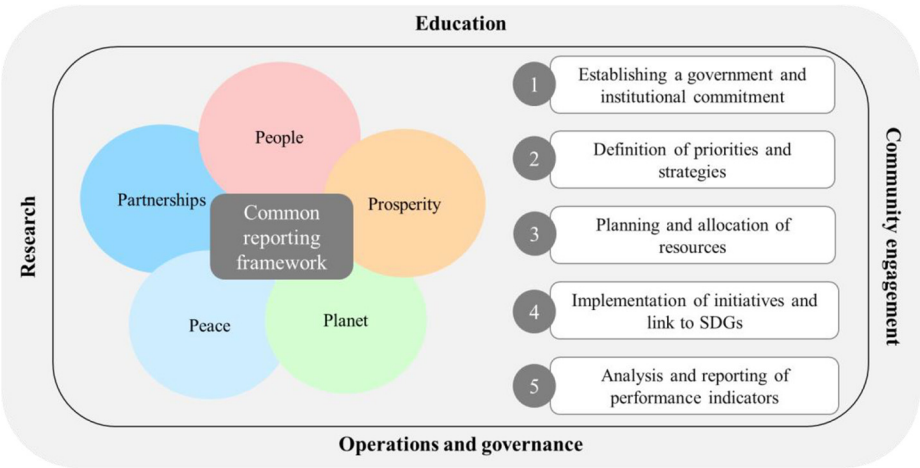


Figure 3.
Proposal of a five-
pillar approach for
the implementation
and reporting of the
SDGs

Source: Own elaboration based on Leal Filho *et al.* (2021)

Thus, this research contributes to advance the knowledge about SDG performance and reporting that had been unexplored in Portugal, while developing a methodology that can be easily replicated in other countries. The reporting assessment framework presented, through a structure composed of simple concepts – but relevant in the context of HEIs – can assist in the identification of important topics to homogenize the information to be reported and improve the comparability of SDG performance among HEIs.

In practical terms, this work can stimulate more HEIs to adhere to the THE_IR and align their practices according to their assessment requirements, becoming more transparent and internationally recognized. On the other hand, at the level of education, the results of this study can help in the development of new courses offering or the readjustment of current programmes to meet the SDGs, especially in themes that have been undervalued (e.g. environmental issues). From a social perspective, participating in sustainability rankings such as the THE_IR and increasing the level of SDG reporting translates into a greater potential for reaching stakeholders, increased awareness and the promotion of more sustainable behaviours within the academic community. Finally, regarding public policies, new directives and legislation could also be considered to stimulate and align the non-financial reporting of HEIs.

However, this work has some limitations. One of them is the fact that the main reports publicly available on the HEI websites were used, and the names and depth of information are not homogeneous across institutions. In addition, the reports analysed referred to the year 2020 and may have been affected by the COVID-19 pandemic. On the other hand, we did not consider the existence of sustainability reports where there could be more reference to SDG topics, because not all Portuguese HEIs present in the THE_IR 2022 publish sustainability reports.

Another limitation is the subjectivity in the choice of key topics in the reporting assessment framework. Given the scarcity of structures adapted to the higher education context, it has been defined the topics with the most appropriate keywords for HEIs. However, this framework is not a finalized and unalterable proposal but rather a useful starting point for the standardization and alignment of information provided by HEIs. Thus, other key indicators and topics could be added to this framework to complete the process of developing a universal model, which requires more empirical research.

The content analysis method has a subjective and limited aspect dependent on the availability of information regarding the searched keywords, which results in an additional constraint. Therefore, new opportunities for future research could emerge to complement analysis of the content of reports, such as the application of surveys to HEI presidents or rectors.

Note

1. To ensure anonymity, HEIs will not be identified. Additional information can be found in THE_IR (2022a).

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SDG	Key topics
1 End poverty	1. Eradicate/end poverty 2. Aid/support/creation of opportunities for the poorest/vulnerable/needily in poverty or social exclusion 3. Support/grants for students with difficulties/financial/economic needs/economically in need 4. Reception/support for students from underdeveloped/developing countries or from the community of Portuguese-speaking countries or Portuguese-speaking African countries 5. Supporting local business/start-ups or local development programs
2 Zero hunger	1. Eradicate/end hunger 2. Provision of sustainable/healthy/quality meals/food options 3. Promoting food security or healthy eating habits/behaviours 4. Collection/donation/delivery of food or cooperation with the "Food Bank against Hunger". 5. Purchases of local food or from sustainable sources
3 Good health and well-being	1. Health promotion and/or quality health and/or well-being 2. Promoting healthy lifestyles or improving campus quality/living conditions 3. Promotion of sport/activity or exercise/sport 4. Existence of services/monitoring/medical/psychological/psychiatric support 5. Collaboration/partnerships with institutions/units/health centres and/or hospitals
4 Quality education	1. Promotion of quality/innovative/qualified/diversified/excellent education or training 2. Promotion of lifelong learning/training/education 3. Improvement of teaching/learning quality or adequacy/satisfaction with teaching/learning resources 4. Availability of resources open to the community (library, computers, educational content or open classes) 5. Promotion of workshops/seminars/lectures/webinars/conferences
5 Gender equality	1. Promotion of gender equality 2. Promoting equality between men and women 3. Combating/eliminating inequalities/discrimination/violence against transgender or gender-based people 4. Promotion of women/female representation to study/in leadership/in decision-making positions/publications 5. Maternity/paternity or parenting support/policy
6 Clean water and sanitation	1. Availability of clean water and/or clean/safe/accessible sanitation 2. Promotion of water quality/treatment/care/reuse 3. Reduction of water consumption 4. Water efficiency/sustainability or consumption/efficient/conscious use of water 5. Prevention of water pollution
7 Affordable and clean energy	1. Use of renewable/clean/sustainable/accessible energies/sources of energy 2. Reduction of energy consumption 3. Promoting energy efficiency/sustainability or efficient/conscious energy consumption 4. Carbon reduction (CO₂)/emissions or carbon footprint 5. Making buildings more energetically efficient or optimizing the energy consumption of buildings

(continued)

Table A1.
SDGs reporting
assessment
framework

SDG	Key topics
8 Decent work and economic growth	1. Promotion of worthy/decent work or improvement of working conditions 2. Promotion of medicine/health/safety at work 3. Contribution to economic growth/development/value creation 4. Promoting employability or job creation 5. Support for the inclusion of graduates/students in the labour/professional life market
9 Industry, innovation and infrastructure	1. Collaboration/interaction/knowledge transfer to industry 2. Support/encouragement for spin-offs creation 3. Promoting/supporting innovation/innovative projects/technological development 4. Upgrading/improving the conditions or quality of infrastructure 5. Patent registration/application
10 Reduce inequalities	1. Reducing/combating inequalities or discrimination 2. Promotion of equal opportunities 3. Promotion of inclusion/integration/equity 4. Eliminate/overcome cultural/religious/racial/linguistic barriers 5. Improvement of physical accessibility or facilities accessible to persons with disabilities or special needs or reduced mobility
11 Sustainable cities and communities	1. Promotion of sustainable/resilient cities/communities or urban rehabilitation/development 2. Promotion of sustainable/soft mobility or use of bicycles/electric vehicles 3. Enhancement/preservation/safeguarding of heritage or arts 4. Support or provision of accommodation/school residence 5. Remote working/remote working permission
12 Responsible consumption and production	1. Sustainable production/consumption or sustainable/rational/efficient resources management 2. Reduced use/consumption of plastic or paper 3. Reduction/recovery of waste or increase/incentive for recycling/reuse 4. Reduction of food waste or resources or processes 5. Publication of a sustainability report
13 Climate action	1. Climate action 2. Combating climate change 3. Low carbon energy use 4. Carbon neutral university/institute or reach of carbon neutrality 5. Reduction/mitigation of greenhouse gas (GHG) emissions
14 Life below water	1. Protect marine/underwater life 2. Conservation/sustainable use/protection of oceans/seas/lakes/rivers/marine resources 3. Protection/preservation of marine/aquatic ecosystems or coastal/oceanic resources 4. Reducing or combating marine waste/pollution 5. Promoting sustainable fishing or combating overfishing
15 Life on land	1. Protecting earthly life or life on earth 2. Sustainable development or maintenance of land/forests/green areas or spaces 3. Conservation/promotion of biodiversity or preservation of species/fauna/flora or fight against biodiversity loss 4. Planting of trees 5. Reduction/collection of plastic waste/hazardous waste/urban solid waste

Table A1. (continued)

SDG	Key topics
16 Peace, justice and strong institutions	1. Promoting peace or justice or freedom 2. Participation/involvement with the academic/student association or stakeholders 3. Combating/preventing corruption/bribery or risk management plan for corruption and related infringements 4. Promoting transparency of processes/procedures/management 5. Collaboration with the government/city council/public or local administration
17 Partnerships for the goals	1. Partnerships for the implementation of the objectives/SDGs 2. Promoting dialogue/alliances/collaborations for sustainability/SDGs 3. International collaboration/partnerships or internationalization of the institution 4. Sustainability or SDGs education/programmes/courses 5. Publication of an SDGs report/agenda

Source: Own elaboration based in [Nardo et al. \(2021\)](#) and [THE_IR \(2022b\)](#)

Table A1.

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